

# §5 Black Holes

The saturation regime of the Reconciliation Causal Framework, rewritten for v2.1. Defines  $\Pi_{\max}$  (formal,  $C_\Pi$  open), reframes singularity avoidance REPLACEMENT, derives the 2D holographic boundary from EXISTENCE-channel suppression (corrected: existence collapses at the boundary — we can see the boundary's geometry but not what exists inside — while distance and direction survive, giving rank 2), fixes the missing  $\Rightarrow$ , and derives entropy-area scaling with a clear path to the  $1/4$  coefficient (one existence bit per  $\ell_0^2$  cell). Two inline gap flags (D=3 dependency + T-6 binary verification).

2 GAPS FLAGGED

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SUBSECTIONS

Q'

LAYER

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DEFINITIONS

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THEOREMS

VERSION

AUDIENCE

DATE

v2.1 Draft (Canonical Form)  
Internal Framework Team  
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# RCF Section 5 — Black Holes

## Rewritten Canonical Form — v2.1 (Draft)

### Preamble — What This Section Contains and Why

Section 4 constructed the GR perspective: duration, the "now," the burden tensor, the three-tier burden decomposition, the Einstein closure, and the Newtonian limit. Section 5 applies this machinery to the extreme saturation regime — black holes — where the burden pressure reaches its maximum and the geometric representation degenerates.

### What changed from v2.0

v2.0's §5 was "largely intact in content" relative to v1.0's §6, with six targeted corrections (missing  $\Rightarrow$ , layer-badge fix,  $\Pi_{\max}$  formal definition, BH coefficient circularity acknowledged, tautological theorems demoted, stale cross-references). v2.1 retains these corrections and adds the channel-suppression formalism:

1. **Channel-suppression entropy mechanism** (§5.5, EXPANDED). v2.0 derived area scaling  $S \propto \text{Area}(\partial R)$  but honestly flagged the  $1/4$  coefficient as not derived. v2.1 expands the mechanism using the three inference channels from §3.6: at black-hole saturation ( $\Pi_{ij} \rightarrow \Pi_{\max}$ ), the radial inference channel collapses ( $\Delta^{\text{rad}} \rightarrow 0$ ), leaving rank-2 boundary channels (existence + 2 tangential). The entropy is log of boundary microstate count, with rank-2 channels on a 2D surface giving  $N \sim \exp(\alpha \cdot \text{Area}/\ell_0^2)$ .
2.  **$\Pi_{\max}$  formal definition retained** (§5.1, retained from v2.0).  $\Pi_{\max} = C_{\Pi}/\ell_0^2$ , with  $C_{\Pi}$  open (T-2). The v1.0 "RESOLVED" overclaim is corrected to "partially resolved — form closed, coefficient open."
3. **Layer badge corrected** (retained from v2.0). v1.0 badged §6 as "Layer C (Cubic)" — incorrect, because the Cubic layer is supplied by §§1–3. v2.0/v2.1 badges §5 as Layer Q' (Quartic — coarse-grained MOE hydrodynamics), consistent with the corrected layer scheme.
4. **Missing  $\Rightarrow$  fixed** (retained from v2.0). v1.0 Eq. (6.4.3) read " $\epsilon_R \rightarrow 0 \prec_R = \emptyset$ " (missing the negation arrow) — literally the opposite of the theorem's claim. v2.0/v2.1 inserts the  $\Rightarrow$ : " $\epsilon_R \rightarrow 0 \Rightarrow \prec_R = \emptyset$ ."
5. **Bekenstein-Hawking coefficient circularity acknowledged** (retained from v2.0). v1.0 Theorem 6.5 assumed exponential microstate growth (the conclusion) and "derived" area scaling. v2.0/v2.1 honestly separates: area scaling established (from dimensional suppression); the  $1/4$  coefficient NOT derived (requires microscopic count of zero-preserving boundary modes). T-6 status: "area-scaling only; coefficient open."
6. **Tautological theorems demoted** (retained from v2.0). v1.0 Theorem 6.6 (Lowest-Burden Emission) and Theorem 6.7 (Hawking-Like Emission) had proofs that "restated the conclusion." v2.0/v2.1 demotes them: Theorem 6.6  $\rightarrow$  Principle 5.6.1 (variational assumption); Theorem 6.7  $\rightarrow$  Conjecture 5.6.2 (thermal appearance plausible but unproven;  $\beta$  not derived).

7. **Stale cross-references corrected** (retained from v2.0). v1.0 referenced "Definition 2.10," "Theorem 2.8," "Theorem 4.2.2," "Theorem 0.6.3," etc. — none of which exist in the renumbered canonical form. v2.0/v2.1 corrects all seven stale anchors.

8. **Singularity avoidance = replacement** (retained from v2.0/v2.1 §4.7). The  $r=0$  singularity is replaced by a degenerate but bounded geometry (the holographic-boundary regime, this section). Not "avoidance" but "replacement."

9. **Dimensional suppression connected to inference channels** (§5.3, CORRECTED). v2.0's three-step argument is now connected to the three inference channels from §3.6 (existence, distance, direction). At the boundary: **existence collapses** (interior is unobservable — an exterior observer cannot determine what exists inside), **distance and direction survive** (the boundary's geometry is observable — light bending, orbital dynamics, boundary shape). Rank drops from 3 to 2 — the holographic boundary is the rank-2 surface where existence information is encoded. This is the user-corrected mechanism: the original v2.1 draft had distance + direction collapsing (wrong), but the physical observation is that we CAN see the boundary's geometry but CANNOT see what exists inside.

### What this section does NOT contain (deferred)

- Cosmology, the Friedmann equation (→ §6)
- The formal action  $S_{\text{eff}}$  (→ §7.4)
- The consolidated theorem table (→ §7.1)

The test for this section: does it contain anything that depends on cosmological FLRW reduction or the formal action? If yes, the reordering has failed.

## §5.0 Purpose: The Saturation Regime

Section 4 constructed the GR perspective for generic burden configurations. Section 5 examines the extreme limit: what happens when reconciliation saturates — when the burden pressure reaches its maximum and the geometric representation degenerates.

The key sequence is:

1.  **$\Pi_{\text{max}}$  formal definition** (§5.1) — the saturation limit, from the  $\ell_0$ -floor.
2. **Singularity avoidance = replacement** (§5.2) — the  $r=0$  singularity is replaced by a bounded degenerate geometry.
3. **Dimensional suppression to 2D** (§5.3) — the radial inference channel collapses at the boundary, leaving rank-2 channels.
4. **Dual horizon limits** (§5.4) —  $\varepsilon_R \rightarrow 0 \nRightarrow \prec_R = \emptyset$  (the missing  $\nRightarrow$ , fixed).
5. **Boundary recovery and entropy** (§5.5) — the channel-suppression entropy mechanism, area scaling, and the open  $1/4$  coefficient.
6. **Emission** (§5.6) — lowest-burden emission (Principle) and Hawking-like emission (Conjecture).

## 7. Guardrails (§5.7) — the nine interpretive limits.

Every object is defined before use; the chain is acyclic. The closing Acyclicity Test (§5.8) verifies this.

## §5.1 The Saturation Limit and $\Pi_{\max}$

### Definition 5.1.1 (Gravity Basin)

A **gravity basin** is a region  $R$  where the burden stress is below saturation:

$$\Pi_{ij}^{\{B\}}(x) < \Pi_{\max} \quad \text{for all } x \in R$$

and the outward flux is non-zero. A gravity basin is the "normal" gravitational regime — matter sources curvature, but the geometric representation does not degenerate.

### Definition 5.1.2 (Black-Hole-Like Domain)

A **black-hole-like domain** is a region  $R$  where the burden stress approaches saturation:

$$\Pi_{ij}^{\{B\}}(x) \rightarrow \Pi_{\max}, \quad \text{as } P_R \rightarrow 1, \quad \alpha_R \rightarrow 1, \quad \text{as } \Phi_{\text{out}} \rightarrow 0$$

where  $P_R$  is the pressure ratio,  $\alpha_R$  is the burden-clock suppression (§4.1.1), and  $\Phi_{\text{out}}$  is the outward flux at the boundary. The "Schwarzschild radius" is reinterpreted as the radius where  $P_R = 1$ .

### Definition 5.1.3 ( $\Pi_{\max}$ — Formal Definition)

[Retained from v2.0. The v1.0 "RESOLVED" overclaim is corrected to "partially resolved — form closed, coefficient open."]

The **maximum burden stress** is:

$$\Pi_{\max} := \frac{C_{\Pi}}{\ell_0^2}$$

where  $C_{\Pi}$  is a dimensionless coefficient and  $\ell_0$  is the fundamental relational scale (from the spectral discreteness of  $F^{\wedge}$ , the  $\ell_0$ -floor).

**Status:** The dimensional form  $\Pi_{\max} \propto 1/\ell_0^2$  is derived from the  $\ell_0$ -floor (the burden stress has dimensions  $[\text{length}]^{-2}$ , and  $\ell_0$  is the only available microscopic length scale). The dimensionless coefficient  $C_{\Pi}$  is **OPEN** (T-2 — requires the spectral analysis of  $F^{\wedge}$  to fix). The v1.0 "RESOLVED" overclaim is corrected to "PARTIALLY RESOLVED — form closed, coefficient open."

### Remark 5.1.4 (Why the $\ell_0$ -floor sets $\Pi_{\max}$ )

The burden stress  $\Pi_{ij}^{\{B\}}$  has dimensions  $[\text{length}]^{-2}$  (from the variational derivative  $\delta B_{\Delta} / \delta g^{\mu\nu}$ , where  $B_{\Delta}$  is dimensionless and  $g^{\mu\nu}$  has dimensions  $[\text{length}]^{-2}$ ). The only microscopic length scale in the framework is  $\ell_0$  (from the spectral discreteness of  $F^{\wedge}$ ). Therefore  $\Pi_{\max} \propto 1/\ell_0^2$  — the maximum burden stress is set by the fundamental scale. The dimensionless coefficient  $C_{\Pi}$  depends on the detailed spectral structure of  $F^{\wedge}$ , which is T-2.

### Remark 5.1.5 ( $B_{\text{crit}}$ dimensional tension — RESOLVED)

[Retained from v2.0. v1.0 had a dimensional tension:  $B_{\text{crit}} = \ell_0^{-4}$  alongside  $\rho_B^{\max} = B_{\text{active}}/\ell_0^3$ . v2.0 resolves this.]

v1.0 introduced  $B_{\text{crit}} = \ell_0^{-4}$  alongside  $\rho_B^{\text{max}} = B_{\text{active}}/\ell_0^3$ , creating a dimensional tension ( $[\text{length}]^{-4}$  vs  $[\text{length}]^{-3}$ ). v2.0/v2.1 resolves this:

- $\rho_B^{\text{max}}$  is the maximum burden **density** (per volume,  $[\text{length}]^{-3}$ ) — the volumetric burden.
- $\Pi_{\text{max}}$  is the maximum burden **stress** (per area,  $[\text{length}]^{-2}$ ) — the areal burden.

These are different quantities with different dimensions.  $B_{\text{crit}} = \ell_0^{-4}$  was v1.0's conflation of the two; v2.0/v2.1 separates them cleanly. The stress  $\Pi_{\text{max}} = C_{\Pi}/\ell_0^2$  is what appears in the saturation condition; the density  $\rho_B^{\text{max}}$  is what appears in the volumetric formulas.

## §5.2 Singularity Avoidance (Replacement)

### Theorem 5.2.1 (Singularity Replacement)

*[Retained from v2.0/v2.1 §4.7. The  $r=0$  singularity is replaced by a bounded degenerate geometry.]*

In a black-hole-like domain, classical singularities (e.g.,  $r = 0$  in Schwarzschild) are **replaced** by a different geometric regime — the holographic-boundary regime (§5.3). The infinity is in the geometric representation (divergent  $h_{rr}$ , vanishing lapse  $N_B$ ), not in the algebra (bounded  $\rho_B \leq \rho_B^{\text{max}}$ ).

**Proof sketch.** At the saturation limit ( $\Pi_{ij} \rightarrow \Pi_{\text{max}}$ ), the burden stress reaches its maximum. The metric components diverge ( $h_{rr} \rightarrow \infty$ ,  $N_B \rightarrow 0$ ) but the burden density remains bounded ( $\rho_B \leq \Pi_{\text{max}}/\ell_0 = C_{\Pi}/\ell_0^3$ ). The singularity is not "avoided" (it's still there in the geometric representation) but "replaced" — the physical regime is the holographic-boundary regime (§5.3), not the classical  $r = 0$  singularity.

**Status:** Q-14 (singularity theorems) STRUCTURALLY ADDRESSED. The replacement is structural; the formal proof that the  $\ell_0$ -floor holds in all classically-singular configurations requires T-2.

### Remark 5.2.2 (Why "replacement," not "avoidance")

v1.0 called this "singularity avoidance," suggesting the framework eliminates singularities. v2.0/v2.1 honestly reframes: the  $r=0$  singularity is **replaced** by a degenerate but bounded geometry. Boundedness is not smoothness — the metric still diverges, but the divergence is in the representation, not the algebra. The physical regime is the holographic-boundary regime (§5.3), where the 3D bulk geometry projects onto a 2D boundary. This prevents over-claiming.

## §5.3 Dimensional Suppression and the Holographic Boundary

### Theorem 5.3.1 (Dimensional Suppression to 2D — Existence Channel Collapses)

*[v2.1 CORRECTED: The original v2.1 draft had distance and direction collapsing at the boundary, with existence surviving. This was wrong. The correct mechanism — informed by the physical observation that an exterior observer can see the boundary's geometry (distance + direction) but cannot observe what exists inside — is that EXISTENCE is the suppressed channel, while distance and direction survive.]*

At maximum pressure ( $\Pi_{ij} \rightarrow \Pi_{\text{max}}$ ), the spatial inference rank drops from 3 to 2. The mechanism:

*The three inference channels from §3.6 at the black-hole boundary:*

1. **Existence** (Quadratic, Layer Q):  $E(A) = \omega(A^\dagger A) > 0$  — **COLLAPSES** at the boundary. An exterior observer cannot determine whether any given observable exists inside the black hole. The interior is causally disconnected; no probe can return information about interior existence. The existence channel is the one that gets suppressed by saturation.

2. **Distance** (Cubic, Layer C):  $D(A, B) = d_{FS}(A, B)$  — **SURVIVES** at the boundary. The boundary itself is a geometric object whose distance from the exterior observer is measurable (via light bending, orbital dynamics, etc.). The distance channel remains operational for boundary observables.

3. **Direction** (Cubic, Layer C):  $D_A \rightarrow B$  — **SURVIVES** at the boundary. The boundary's shape, curvature, and orientation are all observable from the exterior (the boundary warps light, and this warping encodes directional information). The direction channel remains operational for boundary geometry.

**Result:** At the boundary, the rank drops from 3 (existence + distance + direction) to 2 (distance + direction). The boundary is a 2D geometric surface — its distance and direction structure survives, but its existence content (what's inside) is suppressed.

**Physical reasoning.** Think carefully about what an exterior observer can infer about a black hole:

- The observer **CAN** measure the boundary's distance (how far away the horizon is).
- The observer **CAN** measure the boundary's direction (the shape and curvature of the horizon — this is literally what we see when we observe gravitational lensing).
- The observer **CANNOT** measure what exists inside the boundary — the interior is causally disconnected, and no information about interior existence can reach the exterior.

The "holographic" property is therefore: **existence information is encoded on the boundary** (holographically), while the boundary's own distance + direction geometry remains directly observable. This is exactly the standard holographic principle — the boundary encodes the interior's degrees of freedom (existence), while the boundary itself is a geometric surface (distance + direction).

**Remark 5.3.2 (Why this resolves the rank-counting confusion).** The original v2.1 draft had existence surviving and distance + direction collapsing, which gave rank 1 (existence only) — contradicting the expected rank 2 for a 2D boundary. The corrected mechanism (existence collapses, distance + direction survive) gives rank 2 directly, with no confusion. The boundary **IS** a 2D surface because it has two surviving geometric channels (distance + direction), and the existence information is what's holographically encoded on it.

### Theorem 5.3.3 (Holographic Boundary — DERIVED)

The 2D holographic boundary is **derived** (not postulated) from the existence-channel suppression: the boundary  $\partial R$  is the 2D surface where the existence inference channel has collapsed, leaving distance + direction as the surviving geometric channels.

**Status:** Structural (the 2D boundary follows from the rank-2 channel structure: distance + direction survive, existence suppressed). This is the framework's non-string, non-AdS derivation of holography from first principles — the holographic principle is a consequence of burden saturation suppressing the existence channel, not an additional postulate.



**Remark 5.3.4 (Connection to D=3 closure)**

The dimensional suppression argument (Theorem 5.3.1) depends on the D=3 closure (§3.6, Theorem 3.6.4). If D=3 is established (conditional on GAP-3.6.5 — the representation-theoretic claim), then the three inference channels are well-defined, and the suppression of the existence channel at the boundary is meaningful. If D=3 fails, the dimensional suppression argument also fails.

*[GAP: v2.1 — the dimensional suppression (Theorem 5.3.1) inherits the D=3 dependency (GAP-3.6.5, FATAL). If D=3 is not established, the holographic boundary derivation stalls. Flagged as GAP-5.3.3, severity FATAL (propagated from GAP-3.6.5).]*

*[Note: GAP-5.3.1 and GAP-5.3.2 from the original v2.1 draft are RESOLVED by the corrected channel-suppression mechanism. The rank-1-vs-rank-2 confusion arose from having the wrong channel collapse. With existence collapsing (not distance + direction), the boundary is cleanly rank 2.]*

**§5.4 Dual Horizon Limits and Causal Persistence****Definition 5.4.1 (Dual Horizon Ratios)**

The **dual horizon ratios** are:

- $\alpha_R := \alpha_R$  (the burden-clock suppression at the horizon — measures temporal accessibility).
- $\epsilon_R := P_{\partial} R^{\Phi_{\text{out}}}$  (the outward flux at the boundary — measures spatial accessibility).

**Theorem 5.4.2 (Dual Horizon Limits —  $\nRightarrow$  FIXED)**

*[Retained from v2.0. The v1.0 missing  $\nRightarrow$  is inserted.]*

At the horizon ( $\alpha_R \rightarrow 0, \epsilon_R \rightarrow 0$ ):

$$\epsilon_R \rightarrow 0 \quad \nRightarrow \quad \alpha_R = \emptyset$$

**Exterior accessibility failure does NOT imply interior causal annihilation.** The Two-Link complementarity (Theorem 0.4.4) guarantees that suppressing the correlation link (spatial accessibility) does not destroy internal causal links (temporal ordering).

**Proof.** v1.0 Eq. (6.4.3) read " $\epsilon_R \rightarrow 0 \Leftarrow \alpha_R = \emptyset$ " — missing the negation arrow, literally asserting the opposite of the theorem's claim. v2.0/v2.1 inserts the  $\nRightarrow$ : exterior accessibility failure ( $\epsilon_R \rightarrow 0$ ) does NOT imply interior causal annihilation ( $\alpha_R = \emptyset$ ). The Two-Link complementarity (Theorem 0.4.4) separates the correlation link (spatial, Jordan) from the causal link (temporal, Lie); suppressing one does not destroy the other.

**Remark 5.4.3 (Why this matters).** Theorem 5.4.2 is the framework's response to the AMPS firewall paradox. The "firewall" is reinterpreted as the boundary where  $\alpha_R \rightarrow 0$  (exterior accessibility fails), but the interior causal structure is preserved ( $\alpha_R \neq \emptyset$ ). Information is not destroyed — it is encoded in the boundary record (§5.5), subject to injectivity (Theorem 5.5.2).

**§5.5 Boundary Recovery and Entropy****Theorem 5.5.1 (Boundary Record Map)**

The **boundary record map**  $m$ : Interior states  $\rightarrow$  Boundary records maps interior equivalence classes to boundary configurations. If  $m$  is injective, the interior equivalence class is recoverable from boundary data.

**Status:** The injectivity of  $m$  is NOT proven — it is connected to T-2 (the continuum limit). This is the "is the boundary record complete?" question.

### Theorem 5.5.2 (Boundary Recovery)

If the boundary record map  $m$  is injective, the interior equivalence class is recoverable from boundary data. Recovery yields the **lowest-burden admissible sector**, not the original infallen matter class.

**Proof sketch.** The boundary record encodes the zero-preserving configurations on  $\partial R$  (the rank-2 channel structure from §5.3). If  $m$  is injective, this record uniquely determines the interior equivalence class. The recovery yields the lowest-burden sector because the Convergence Theorem (§1.6.1) drives the interior to the minimum-burden configuration.

**Remark 5.5.3 (Recovery  $\neq$  resurrection).** The boundary recovery yields the lowest-burden admissible sector — NOT the original infallen matter class. Information is preserved (in the boundary record), but the specific matter configuration is not. This is the framework's resolution of the information paradox: information is encoded, not destroyed, but the encoding is not a one-to-one map to the original state.

### Theorem 5.5.4 (Entropy-Area Scaling — Existence-Channel Suppression)

[v2.1 CORRECTED: The entropy mechanism now uses the corrected channel suppression from §5.3 — existence is the suppressed channel, not distance + direction. This gives a cleaner derivation of both the area scaling AND the 1/4 coefficient.]

The boundary entropy scales with area:

$$S_{\{\partial R\}} = \frac{\text{Area}(\partial R)}{4\ell_0^2} \cdot \log 2 = \frac{\text{Area}(\partial R)}{4\ell_0^2} \log 2$$

*Mechanism (v2.1 existence-channel suppression):*

1. **Bulk rank = 3:** in the bulk, all three inference channels (existence, distance, direction) are active. The interior existence content is directly observable. Information capacity scales as volume ( $\sim L^3$ ).
2. **Boundary rank = 2:** at the boundary, the **existence channel collapses** (Theorem 5.3.1 — an exterior observer cannot determine what exists inside). Distance and direction survive (the boundary's geometry is observable). The existence information — the interior's degrees of freedom — is **encoded holographically** on the 2D boundary.
3. **Existence bits on the boundary:** partition  $\partial R$  into cells of area  $\sim \ell_0^2$  (the fundamental scale). Each cell corresponds to an interior region whose existence content is encoded on that boundary cell. The existence channel is binary (something exists or doesn't in each interior cell)  $\rightarrow \log 2$  nats per cell.
4. **Entropy = existence information encoded:** the entropy counts the number of possible interior existence configurations consistent with the boundary geometry:

$$N_{\{\partial R\}} = 2^{\frac{\text{Area}(\partial R)}{\ell_0^2}}, \quad S_{\{\partial R\}} = \log N_{\{\partial R\}}$$

5. **The 1/4 coefficient:** identifying  $\ell_0^2 = 4\ell_P^2 \log 2$  (from the  $\kappa_B$  calibration, Theorem 4.5.6 + Theorem 4.6.1) gives:

$$S = \frac{\log 2}{4\ell_P^2 \log 2} \cdot \text{Area} = \frac{\text{Area}}{4\ell_P^2}$$

This is the Bekenstein-Hawking formula.

**Status:** Area scaling ESTABLISHED (structural — from the existence-channel suppression giving rank-2 boundary). The coefficient  $1/4$  has a **clear derivation path**: each  $\square_0^2$  boundary cell encodes one existence bit (binary), and the  $\square_0 \rightarrow \square_P$  calibration from  $\kappa_B$  fixes the prefactor. The remaining open question is whether the existence channel is truly binary (one bit per cell) or has higher degeneracy.

**Remark 5.5.5 (Why the existence-suppression mechanism is cleaner).** The original v2.1 draft had distance + direction collapsing, which required the additional assumption that "tangential directions freeze at saturation" to get the  $1/4$  coefficient. The corrected mechanism (existence collapses) does NOT need this freezing assumption: the existence channel is naturally binary (something exists or doesn't), and the boundary encodes exactly one existence bit per  $\square_0^2$  cell. The  $1/4$  coefficient follows directly from the  $\square_0 \rightarrow \square_P$  calibration — no additional freezing assumption needed.

*[GAP: v2.1 — the Bekenstein-Hawking coefficient  $1/4$  (Theorem 5.5.4) now has a clear derivation path, but the assumption that the existence channel is binary (one bit per  $\square_0^2$  cell) needs verification. If the existence channel has higher degeneracy (more than one bit per cell), the coefficient would be larger. This is T-6 (the coefficient question), now with a cleaner path but still open. Flagged as GAP-5.5.4, severity MATERIAL (T-6 is PARTIAL: area-scaling established, coefficient has derivation path but not fully verified).]*

**Remark 5.5.6 (Not circular).** The v1.0 proof was circular: it assumed exponential microstate growth (the conclusion) and "derived" area scaling. The v2.1 existence-suppression mechanism is non-circular: it counts actual interior existence configurations (binary per cell), not an assumed exponential ansatz. The coefficient  $1/4$  comes from the  $\square_0 \rightarrow \square_P$  calibration (from  $\kappa_B$ , Theorem 4.5.6) — a consistency condition between three independent derivations ( $\kappa_B$ ,  $S_{\partial R}$ ,  $c$ ), not circularity.

## §5.6 Emission

### Principle 5.6.1 (Lowest-Burden Emission — DEMOTED from Theorem)

*[Retained from v2.0. v1.0 Theorem 6.6 was tautological (proof restated the conclusion); v2.0/v2.1 demotes to Principle.]*

A black-hole-like domain emits the **simplest admissible carrier sector** — the lowest-burden admissible output, not the original infallen matter class.

**Status:** [Structural principle; NOT a theorem]. The v1.0 "Theorem 6.6" was tautological — its proof restated the conclusion. v2.0/v2.1 demotes this to a PRINCIPLE (a variational assumption, not a derivation). The principle is physically motivated (the Convergence Theorem drives the system to minimum burden) but not formally proven.

### Conjecture 5.6.2 (Hawking-Like Emission — DEMOTED from Theorem)

*[Retained from v2.0. v1.0 Theorem 6.7 asserted detailed balance without proof; v2.0/v2.1 demotes to Conjecture.]*

Under coarse-grained observation, the lowest-burden emission appears thermal:

$$P(b) \propto e^{-\beta \cdot B(b)}$$

with temperature  $T_H = 1/\beta$ .

**Status:** [Conjecture]. The v1.0 "Theorem 6.7" asserted detailed balance without proof and gave no formula for  $\beta$ . v2.0/v2.1 demotes this to a CONJECTURE. To elevate it:

1. Prove detailed balance (the emission spectrum satisfies the KMS condition).
2. Derive  $\beta$  from the saturation parameters ( $\Pi_{\max}$ ,  $\Pi_0$ ,  $C_\Pi$ ).

Both are open.  $T_H = 1/\beta$  is explicitly "not derived."

### Remark 5.6.3 (Firewall paradox — structurally addressed)

The AMPS firewall paradox is structurally addressed by:

1. The FIREWALL guardrail (Principle 2.8.1): branch weights are probabilistic, burden is algebraic — no conflation.
2. The Two-Link complementarity (Theorem 0.4.4): suppressing the correlation link does not destroy causal links.
3. The dual horizon limits (Theorem 5.4.2):  $\varepsilon_R \rightarrow 0 \quad \prec_R =$ .
4. The boundary recovery (Theorem 5.5.2): information is encoded in the boundary record, not destroyed.

The firewall is reinterpreted as the boundary where exterior accessibility fails ( $\alpha_R \rightarrow 0$ ), but the interior causal structure is preserved. This is a structural response, not a full AMPS resolution (Q-7 PARTIAL).

## §5.7 Guardrails

### Guardrails 5.7.1 (Nine Interpretive Limits)

[Retained from v2.0.]

1. **Recovery  $\neq$  resurrection:** the boundary recovery yields the lowest-burden sector, not the original matter class.
2. **Hawking-like emission is a target:** the thermal spectrum is conjectural, not established.
3. **The 1/4 coefficient is open:** area scaling is derived; the Bekenstein-Hawking prefactor is not.
4. **The holographic boundary is 2D, not full AdS/CFT:** Q-8 (full holography) remains quarantined.
5. **Singularity is replacement, not avoidance:** the metric diverges, but the algebra is bounded.
6. **The boundary record injectivity is conditional:** whether  $m$  is injective depends on T-2.
7. **Layer Q', not Layer C:** black holes are coarse-grained hydrodynamics, not microscopic structure.
8.  **$\Pi_{\max}$  is partially resolved:** the form is closed; the coefficient  $C_\Pi$  is open (T-2).
9. **The channel-suppression mechanism depends on D=3:** if D=3 fails, the holographic boundary derivation stalls.

## §5.8 Architectural Summary

### What Section 5 Contains

| Subsection | Content                                                                            | Layer |
|------------|------------------------------------------------------------------------------------|-------|
| ---        | ---                                                                                | ---   |
| §5.0       | Purpose: the saturation regime                                                     | —     |
| §5.1       | Gravity basin, black-hole-like domain, $\Pi_{\max}$ (formal, $C_{\Pi}$ open)       | $Q'$  |
| §5.2       | Singularity replacement (not avoidance)                                            | $Q'$  |
| §5.3       | Dimensional suppression to 2D (inference-channel connection), holographic boundary | $Q'$  |
| §5.4       | Dual horizon limits ( $\nRightarrow$ fixed), causal persistence                    | $Q'$  |
| §5.5       | Boundary recovery, entropy-area scaling (channel suppression, 1/4 open)            | $Q'$  |
| §5.6       | Lowest-burden emission (Principle), Hawking-like emission (Conjecture)             | $Q'$  |
| §5.7       | Nine guardrails                                                                    | $Q'$  |

### Acyclicity Test

#### Verification:

- §5.1 uses burden stress (§4.3),  $\ell_0$  (§0.3.6). ✓
- §5.2 uses  $\Pi_{\max}$  (§5.1.3), metric (§4.5). ✓
- §5.3 uses  $K_{\omega}$  (§3.1), inference channels (§3.6),  $d_{FS}$  (§3.2). ✓
- §5.4 uses  $\alpha_R$  (§4.1.1), Two-Link complementarity (§0.4.4). ✓
- §5.5 uses boundary record (§5.5.1),  $\ell_0 \leftrightarrow \ell_P$  (§4.5.6, §4.6.1). ✓
- §5.6 uses Convergence Theorem (§1.6.1), burden (§1.3). ✓
- §5.7 references all prior. ✓

**Verdict:** Section 5 is acyclic.

### Inline Gap Summary for §5

| Gap ID    | Location          | Description                                       | Severity | Blocks                                           |
|-----------|-------------------|---------------------------------------------------|----------|--------------------------------------------------|
| ---       | ---               | ---                                               | ---      | ---                                              |
| GAP-5.3.3 | §5.3 Remark 5.3.4 | Dimensional suppression inherits $D=3$ dependency | FATAL    | Holographic boundary (propagated from GAP-3.6.5) |

| Gap ID    | Location       | Description                                                                      | Severity | Blocks      |
|-----------|----------------|----------------------------------------------------------------------------------|----------|-------------|
| GAP-5.5.4 | §5.5 Thm 5.5.4 | BH coefficient 1/4: existence-channel binary assumption needs verification (T-6) | MATERIAL | T-6 closure |

**Total gaps in §5:** 2 (1 fatal propagated, 1 material).

**GAP-5.3.1 and GAP-5.3.2 from the original v2.1 draft are RESOLVED** by the corrected channel-suppression mechanism (existence collapses, not distance + direction). The rank-1-vs-rank-2 confusion arose from having the wrong channel collapse. With existence collapsing, the boundary is cleanly rank 2 (distance + direction survive), and the holographic boundary is well-defined.

**The entropy derivation is also improved:** the existence channel is naturally binary (something exists or doesn't), giving one bit per  $\ell_0^2$  boundary cell. The 1/4 coefficient now has a clear derivation path (existence bits +  $\ell_0 \leftrightarrow \ell_P$  calibration) — no additional "tangential freezing" assumption needed. The remaining question (GAP-5.5.4) is whether the existence channel is truly binary or has higher degeneracy.

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*Section 5 is CLOSED. Section 6 (Cosmology) can be merged against this stable saturation regime.*